The risk posed by *Aedes aegypti* in Cyprus and the wider Mediteranean region

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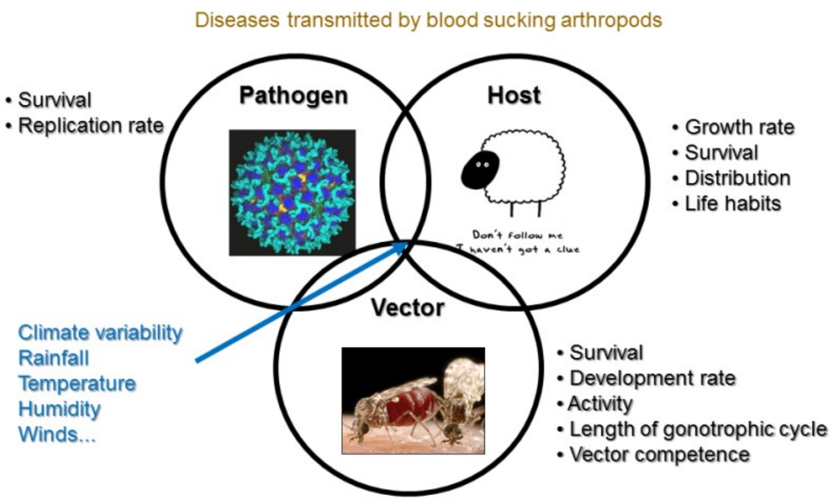
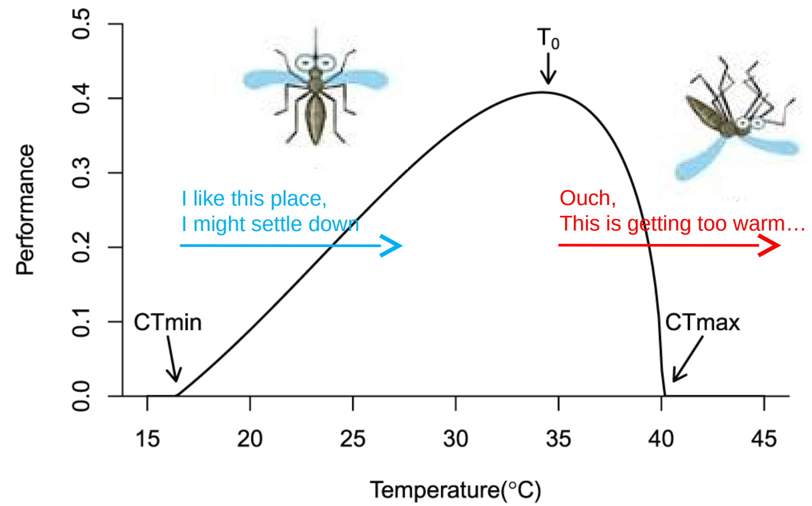
Climate & Vector-borne diseases (VBDs)

VBDs are climate sensitive



Vectorial capacity = F(Tº)

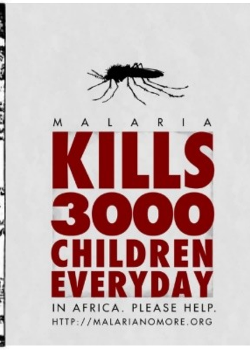


Lafferty KD and Mordecai EA 2016 - [F1000Research 2016, 5:2040]

Modelling the impact of climate variability on VBD burden, development of early warning systems (seasonal to climate change time scales).

Impact of VBD

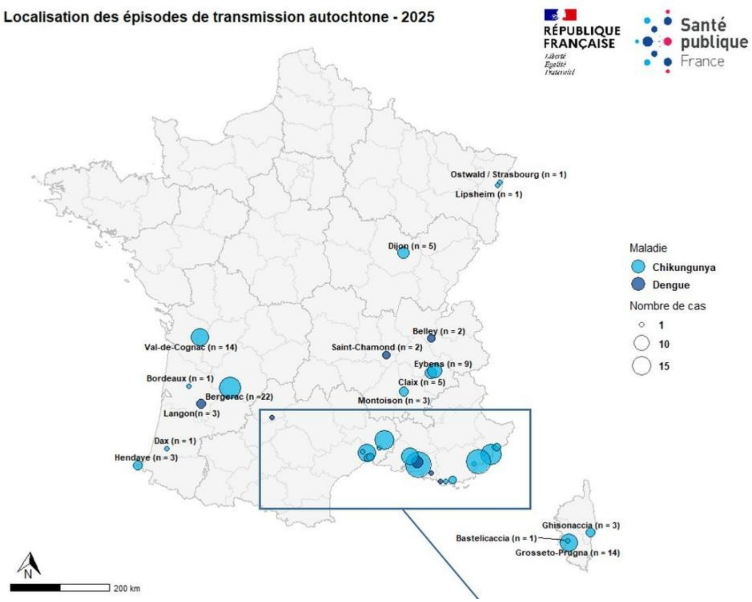
2nd Plague pandemic
14th century



500k deaths/ year in Africa

Zika outbreak, Latin America 2015-16

Yellow Fever in Angola-DRC 2015-16



Distribution of CHIKV & ZIKV **autochthonous** events in France in 2025

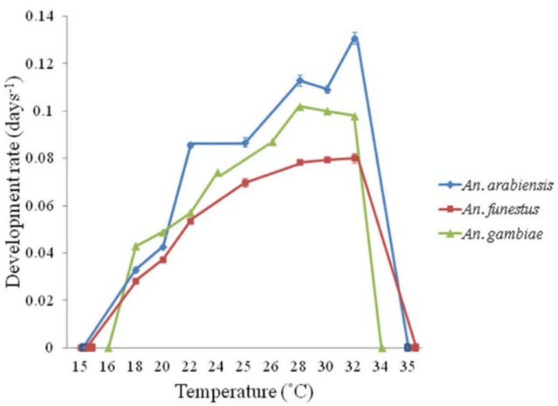
Bluetongue in Europe, 2006

Xylella fastidiosa in Italy

- Large historical impacts of VBDs
- Autochtonous transmission of arboviruses has increased in France & Italy

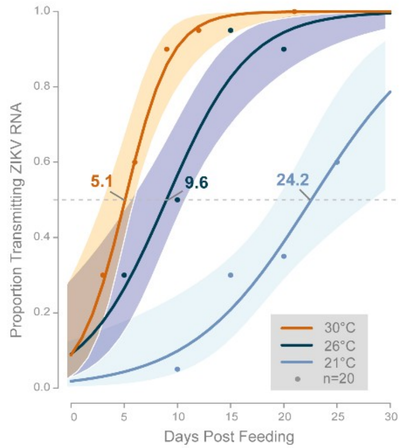
Epidemiological parameters affected by Tº

Development & mortality rate **μ(Tº)**
Example for Anopheles malaria vectors



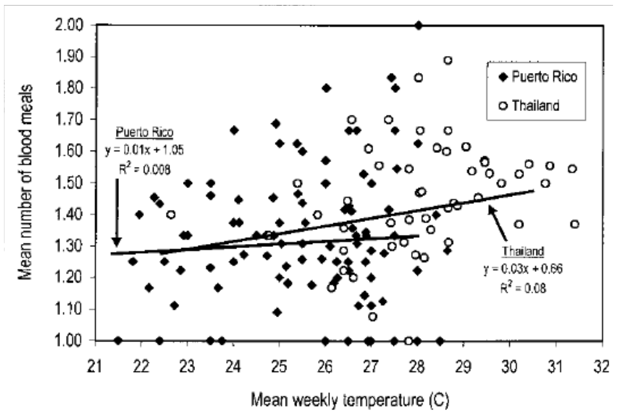
Lyons et al. 2013. *Parasites Vectors* ***6****, 104*

Extrinsic Incubation period **EIP(Tº)**
Example for Ae. aegypti and Zika virus



Winokur et al. 2020. *PLoS Negl Trop Dis* *14(3): e0008047*

Biting rate **a(Tº)**
Example for Ae. aegypti for two sites in Puerto Rico and Thailand

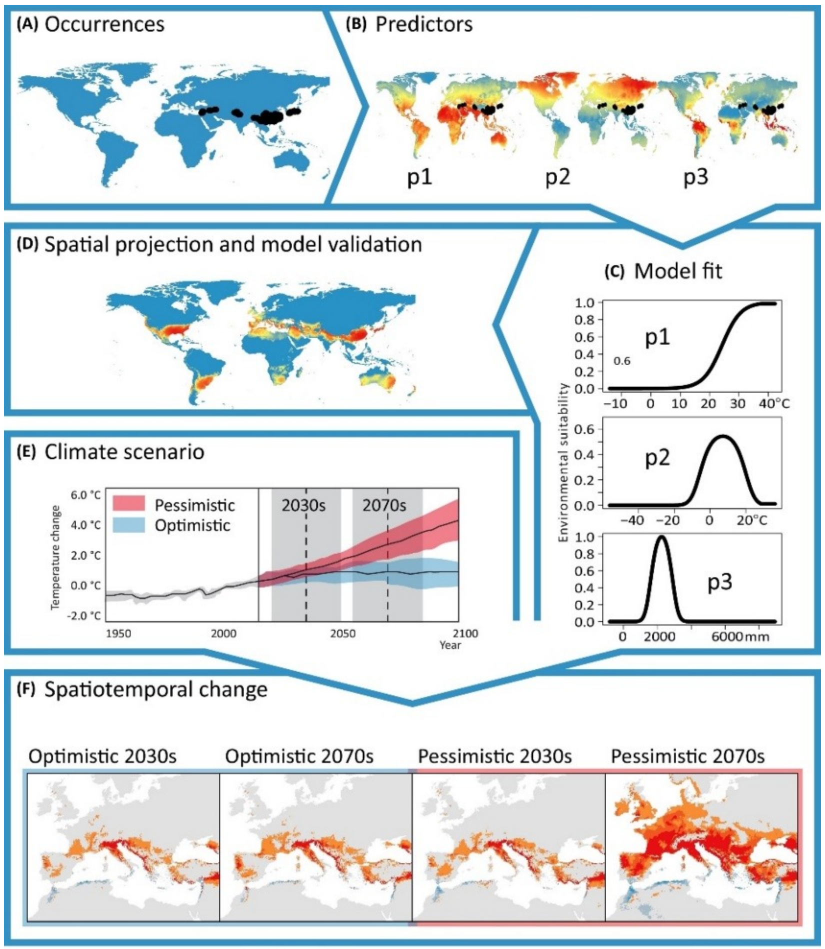


Scott et al. 2000. *J Med Entomol**. 37(1): 89-101*

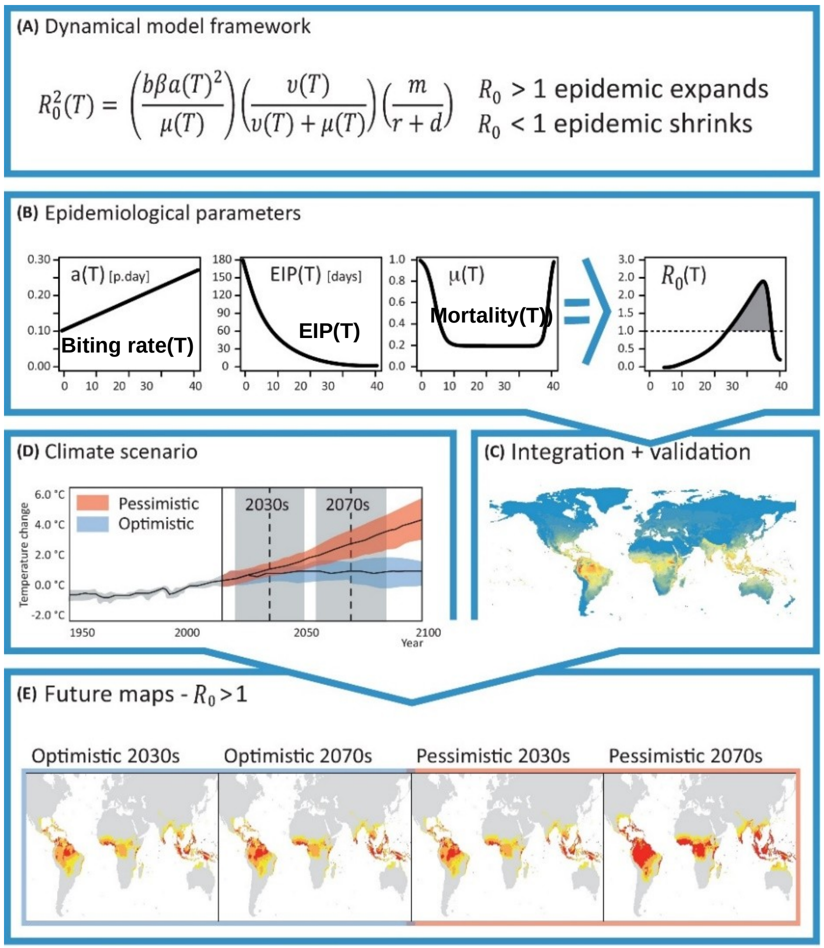
- **Temperature impacts** the biting rate a(Tº), development rate μ(Tº), and the extrinsic incubation period EIP(Tº) that is the time required for the vector to become infectious following an infected bloodmeal.
- **Rainfall** provides breeding sites. Extreme rainfall can lead to flushing of larvae.
- Other factors matter (land use, travel and trade, socio-economic factors, human behaviour, shape of health services…).

Methods to model risk

Statistical models



Mechanistic models



Stat models: Maxent, BRTs, Bayesian models, Mahalanobis distance…

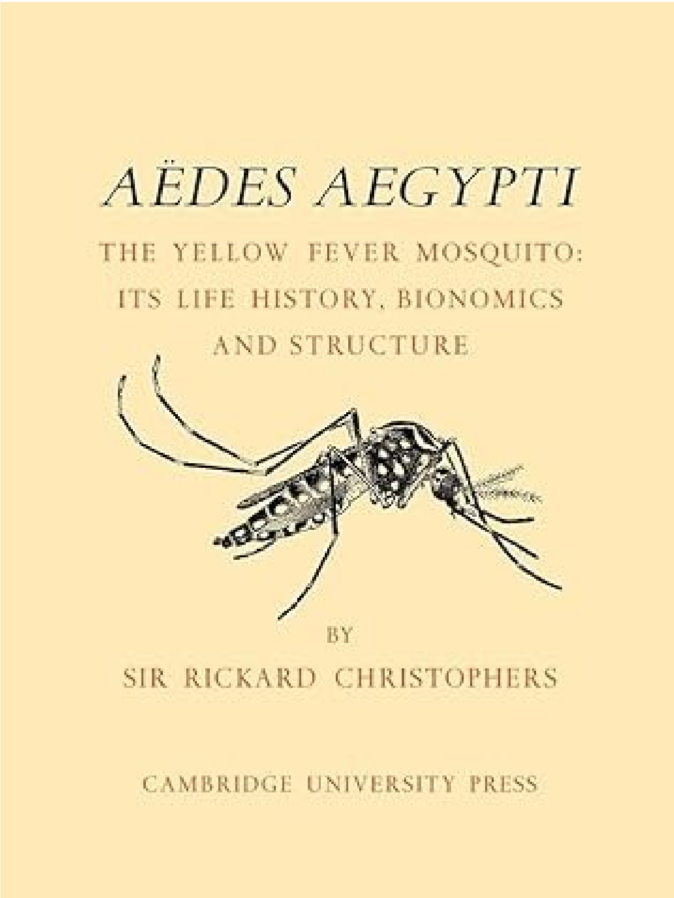
Mechanistic models: SEIR/SIR, R_0, Fuzzy logic, climate envelope…

***Fig. 1 &2** Tjaden et al. (2018). Trends in Parasitology 34(3): 227-245.* http://dx.doi.org/10.1016/j.pt.2017.11.006

1st Vector modelling workshop in Bologna, 19-20 Sep 2024

https://www.vectormodelling.com/Bologna2024/

Key facts about Aedes aegypti



R. Christophers. (1960) *Aëdes Aegypti* (L.) The Yellow Fever Mosquito: Its Life History, Bionomics and Structure. 1st Edition. ISBN-13: 978-0521113021

- *Ae. aegypti* is competent to transmit Yellow fever, dengue, Zika and chikungunya
- It feeds almost exclusively on humans
- It uses man-made water containers to lay eggs and is well adapted to the urban setting
- 40% of the world population is at risk of contracting dengue fever
- It evolved from a sylvatic form (*Ae. aegypti formosus*) to a domestic form *Aedes aegypti aegypti* and originates from Africa – original lineage from Madagascar 7My ago

Historical distribution of *Aedes aegypti*

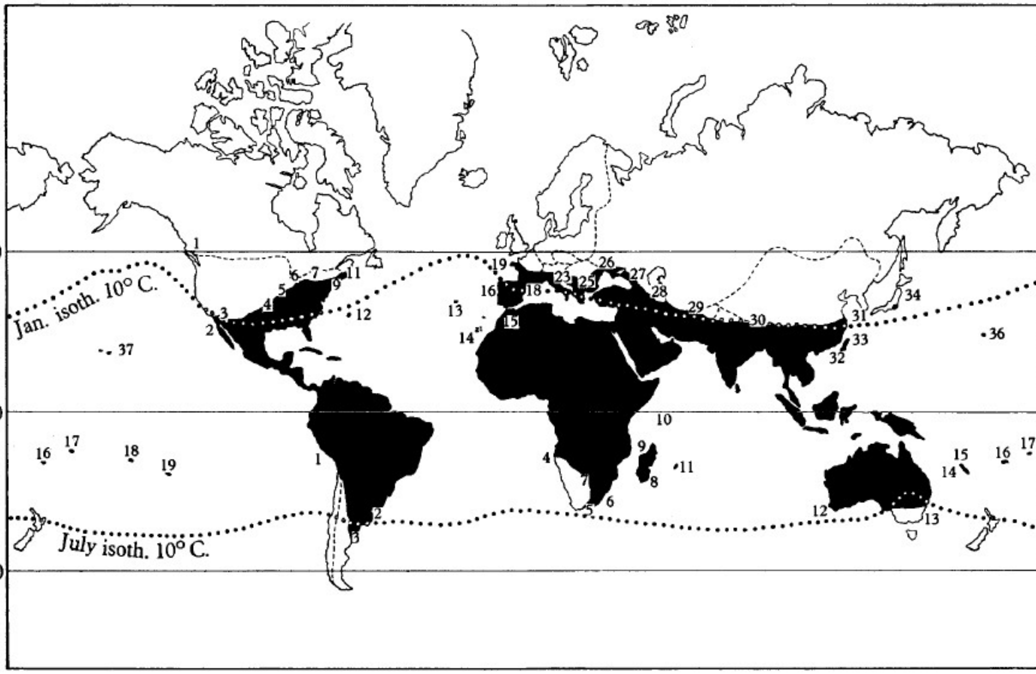


Figure 1. Map showing limits of distribution.

Map of the world showing northern and southern limits of the recorded distribution of *Aëdes aegypti* and isotherms of 10° C.

The numbers are those given to recorded localities as set out in Tables 1 and 2.

Figures in brackets in the tables give the numbers in the list of references to geographical distribution on pp. 50–2.

Table 1. *Recorded northern limits of distribution.**

North America

1	British Columbia (37), 50° N.†
2	Lower California (62), 32° N.
3	Arizona (9), 32° N.
4	Oklahoma (57), 36° N.‡
5	Missouri (40, 62), 38° N.
6	Indiana (40, 62), 40° N.
7	Virginia (40, 62), 38° N.
8	Maryland (40, 62), 39° N.
9	Philadelphia (16), 40° N.†
10	New York (16), 40° 52′ N.†
11	Boston (16), 42° 27′ N.†

Atlantic Islands

12	Bermuda (6), 32° 20′ N.
13	Azores (62), 38° 35′ N.
14	Canary Islands (18, 47), 28° 40′ N.
15	Morocco (Tangier) (17), 35° 42′ N.

Europe

16	Portugal (Badajos) (35), 38° N.
17	Spain (north-west point) (13), 43° N.
18	Spain (Ebro) (27), 41° N.

Europe (*cont.*)

19	France (Brest) (42), 48° 24′ N.
20	France (Dol) (42, 62), 48° 31′ N.
21	France (Bordeaux) (59), 44° 48′ N.
22	Italy (Genoa) (42), 44° 25′ N.
23	Italy (Ravenna) (42), 44° 25′ N.
24	Bosnia (3), 44° N.
25	Macedonia (42), 41° N.
26	Russia (Odessa) (41), 46° 30′ N.‡
27	Russia (Sukkum) (58), 43° N.
28	Russia (Baku) (1), 40° 25′ N.

Asia

29	India (Peshawar) (8), 34° N.
30	India (Assam) (8), 26° N.
31	China (Shanghai) (29), 31° 10′ N.
32	China (Amoy) (30), 24° 40′ N.
33	Formosa (29, 42), 24° N.
34	Japan (Tokyo) (42, 62), 35° 40′ N.
35	Japan (Okayama) (42), 34° 22′ N.

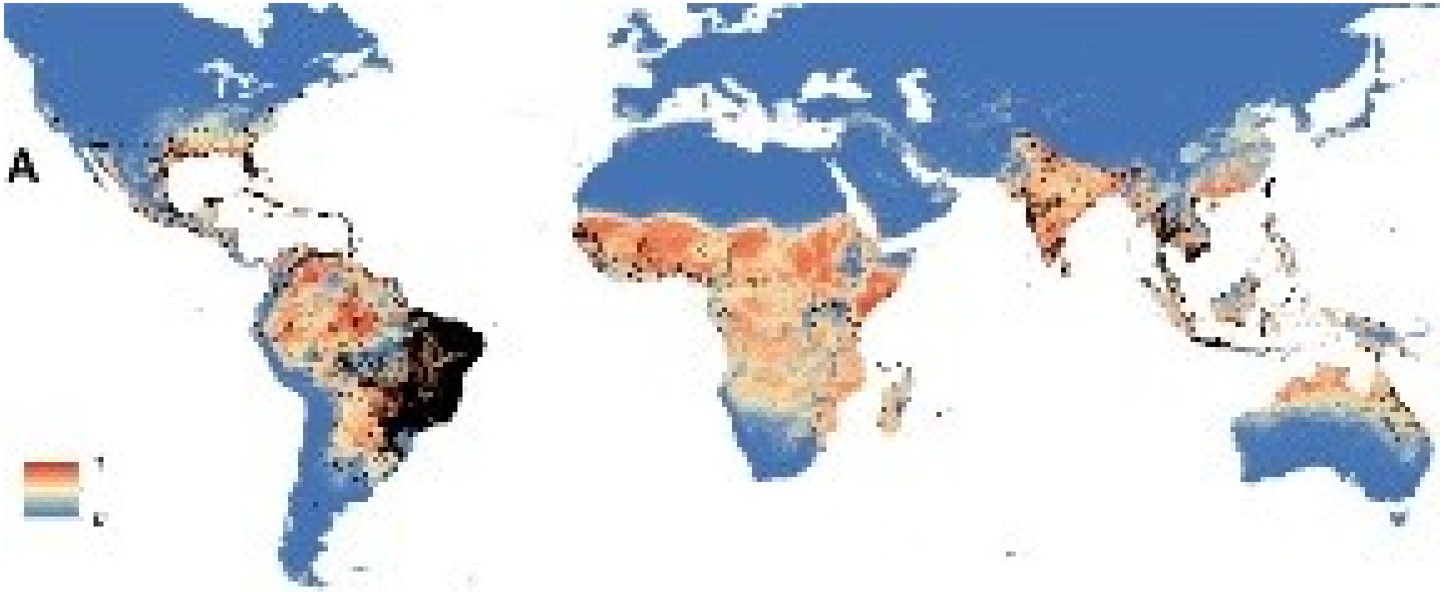
North Pacific Islands

36	Wake Island (56), 20° N.
37	Hawaii (11, 39), 20° N.

R. Christophers. (1960) Aëdes Aegypti (L.) The Yellow Fever Mosquito: Its Life History, Bionomics and Structure. 1st Edition. ISBN-13: 978-0521113021

Distribution of *Ae. aegypti* before control measures were undertaken post WWII.
Southermost occurrence: Buenos Aires; northernmost: British Colombia.
Ae. aegypti traveled by boat and was found in major sea ports…

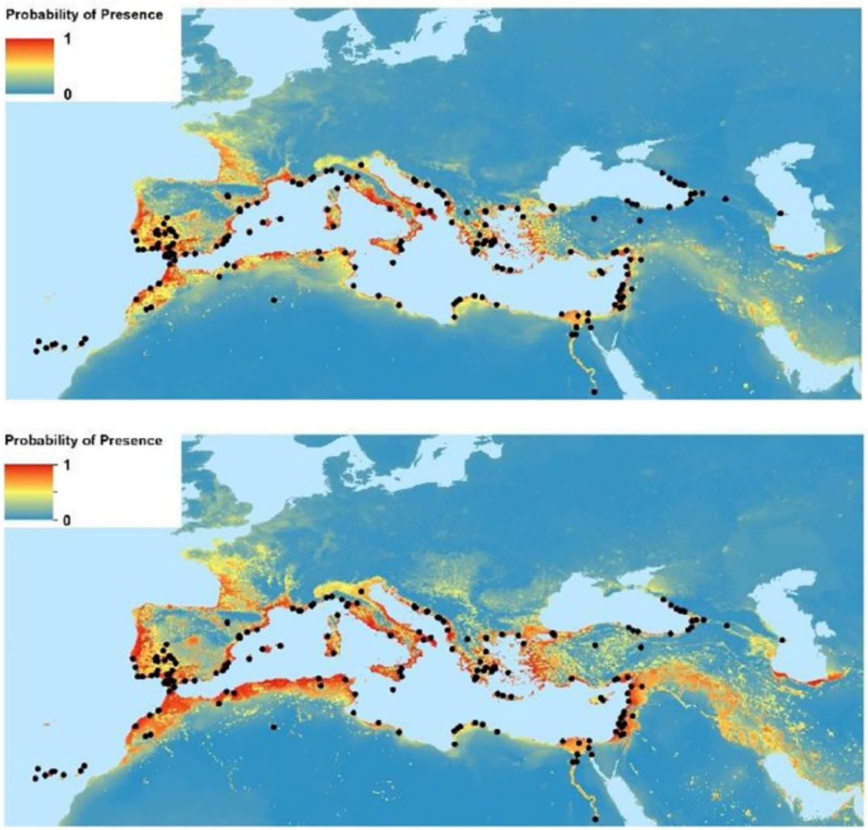
Distribution of *Aedes aegypti in 2014*



Kraemer et al. (2015) The global distribution of the arbovirus vectors *Aedes aegypti* and *Ae. albopictus* eLife 4:e08347.

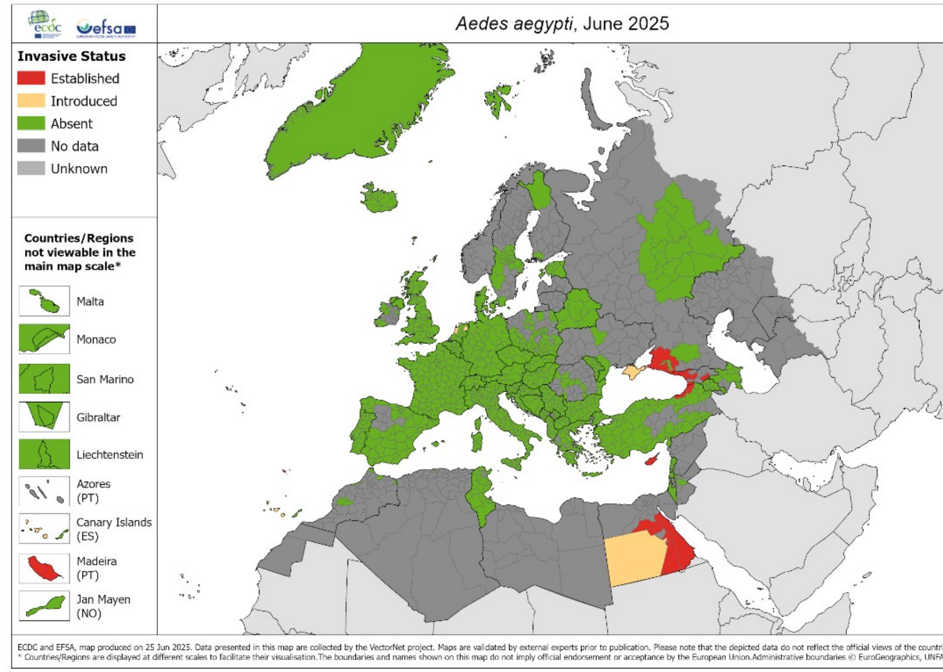
Ae. aegypti has a tropical/semi tropical distribution and is abundant in Latin America, the southern US states, India, Southeast Asia, Australia, and Africa.

Aedes aegypti in Europe and the Mediterranean region



Western Palaearctic region historical 1910 (top) and 2015 (bottom) modelled suitability for *Aedes aegypti* with population in the covariates. Black dots: recorded historical (up to 1955) presence points.

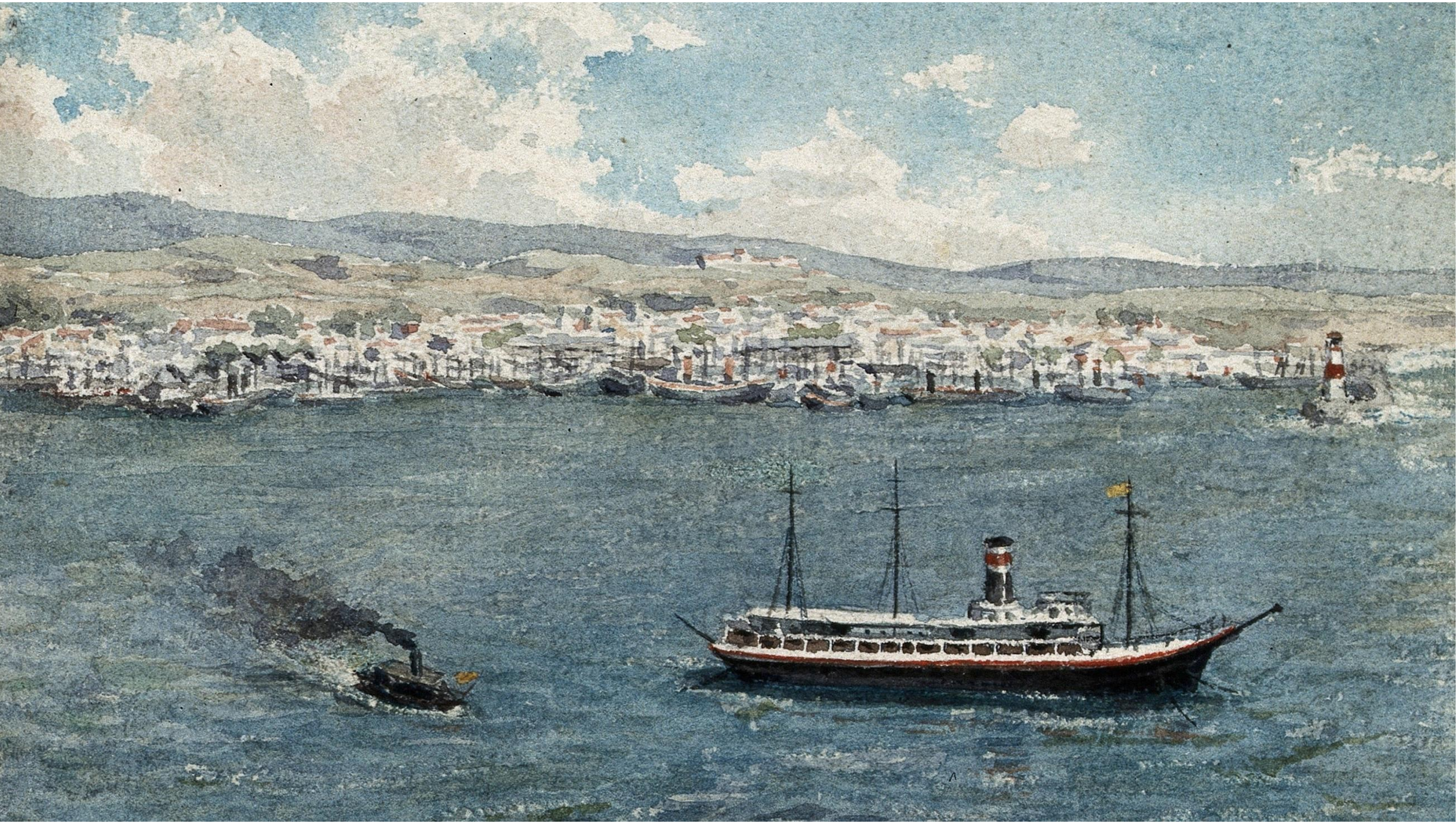
Wint et al. (2022). Sci Total Environ., 847:157566. https://doi.org/10.1016/j.scitotenv.2022.157566



Observed distribution of *Ae. aegypti* in Europe in June 2025

European Centre for Disease Prevention and Control and European Food Safety Authority. Mosquito maps [internet]. Stockholm: ECDC; 2025. Available from: https://ecdc.europa.eu/en/disease-vectors/surveillance-and-disease-data/mosquito-maps

Ae. aegypti was present over southern Europe before control measures were undertaken.
It is now present over the eastern coasts of the Black sea, northwestern Turkey and **Cyprus**.



A yellow quarantine flag, signaling yellow fever, is raised on a ship anchored at sea some distance from a port in this watercolor by E. Schwarz-Lenoir painted some time between 1920 and 1950. CC-BY 4.0 via the Wellcome Collection.

https://www.thinkglobalhealth.org/article/plague-ships-shotgun-mobs-and-saffron-scourge

Ae. aegypti modeling work in Europe and Cyprus

Data & Methods

Data:

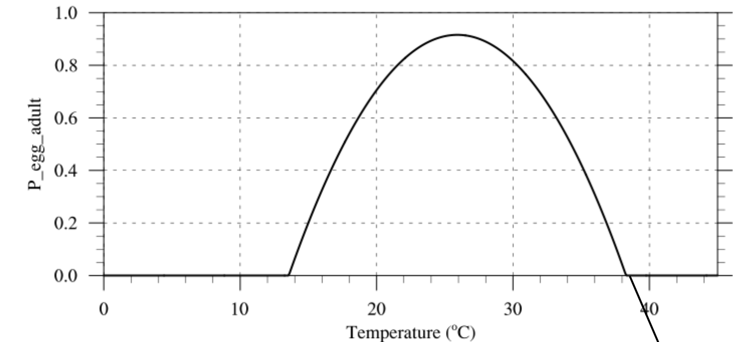
•*Ae. aegypti* historical occurrence data (black dots) derived from Wint et al., 2022
•*Ae. aegypti* ovitrap data for Larnaca, Cyprus (Dr. Marlen Vasquez, Cyprus University of Technology, Limassol) [https://doi.org/10.1101/2024.12.17.628941]
•Lower resolution 25km input climate data derived from EOBS data for Europe [https://doi.org/10.24381/cds.151d3ec6]
•High resolution 1km input climate data (rainfall & temperature) derived from EMO-1 data for Cyprus [https://doi.org/10.2905/0BD84BE4-CEC8-4180-97A6-8B3ADAAC4D26]

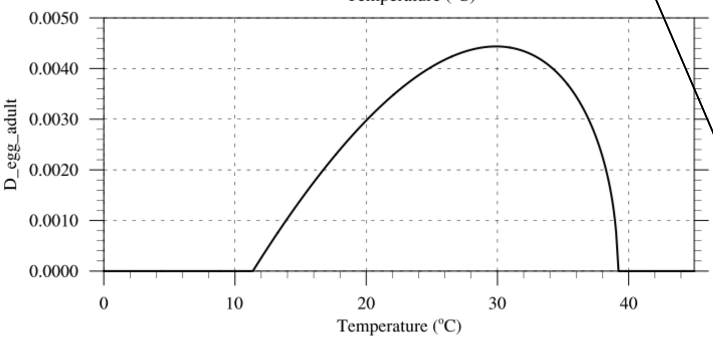
Ae. aegypti models:

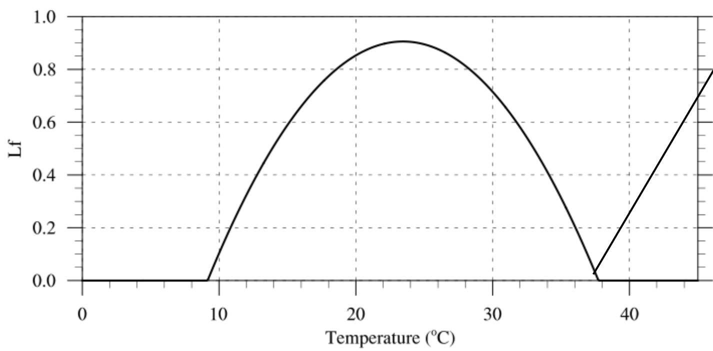
•**Model 1**: Simulated probability of occurrence from Kraemer et al., 2015
•**Model 2**: Mean annual temperature over 10ºC ("old" WHO model & Christophers, 1960)
•**Model 3**: Oviposition rate derived from Yang et al. 2009 [https://doi.org/10.1017/S0950268809002040]
•**Model 4**: VECTRI albopictus model (Garrido Zornoza et al., 2024) with modified adult and larval mortality schemes for *Ae. aegypti* based on Caldwell et al. 2021.

Analysis for Cyprus is mainly based on the *Ae. aegypti* version of the VECTRI model (**Model 4**)

VECTRI *Ae. albopictus* -> *aegypti* prototype







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Garrido Zornoza M., et al. (2024). J. R. Soc. Interface 21:20240319

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The effect of climate change and temperature extremes on *Aedes albopictus* populations: a regional case study for Italy

Miguel Garrido Zornoza, Cyril Caminade and Adrian M. Tompkins

Published: 06 November 2024 https://doi.org/10.1098/rsif.2024.0319

Details References Related Figures

This Issue

Table 3 Fitted thermal responses for *Ae. aegypti* life-history traits.

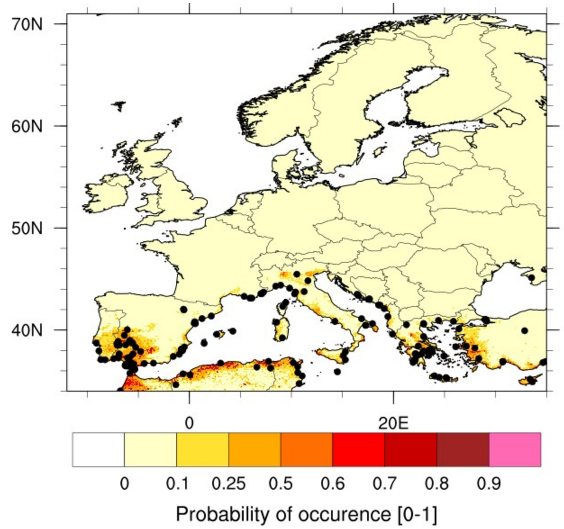
From: Climate predicts geographic and temporal variation in mosquito-borne disease dynamics on two continents

Trait	Definition	Function	c	T_0	T_m
a	Biting rate (day^{-1})	Brière	2.02e^{-04}	13.35	40.08
EFD	Eggs laid per female per day	Brière	8.56e^{-03}	14.58	34.61
pEA	Probability of mosquito egg-to-adult survival	Quadratic	−5.99e^{-03}	13.56	38.29
MDR	Mosquito egg-to-adult development rate (day^{-1})	Brière	7.86e^{-054}	11.36	39.17
Lf	Adult mosquito lifespan (days)	Quadratic	−1.48e^{-01}	9.16	37.73
b	Probability of mosquito infectiousness	Brière	8.49e^{-04}	17.05	35.83
pMI	Probability of mosquito infection	Brière	4.91e^{-04}	12.22	37.46
PDR	Parasite development rate (day^{-1})	Brière	6.56e^{-05}	10.68	45.90

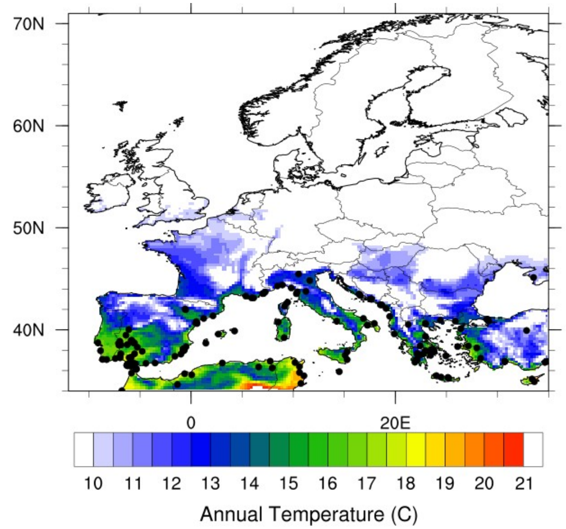
Traits were fit to a Brière [$cT\,(T - T_0)\,(T_m - T)^{\frac{1}{2}}$] or a quadratic [$c\,(T - T_m)\,(T - T_0)$] function where T represents temperature. T_0 and T_m are the critical thermal minimum and maximum, respectively, and c is the rate constant. Thermal responses were fit by ref. [19] and also used in ref. [30]. Parasite development rate was measured as the virus extrinsic incubation rate.

Caldwell. *et al.* (2021). *Nat Commun* **12**, 1233.

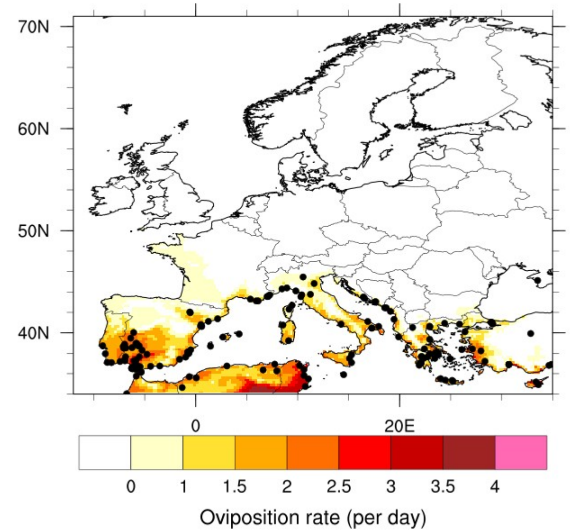
Comparison of *Ae. aegypti* models over Europe



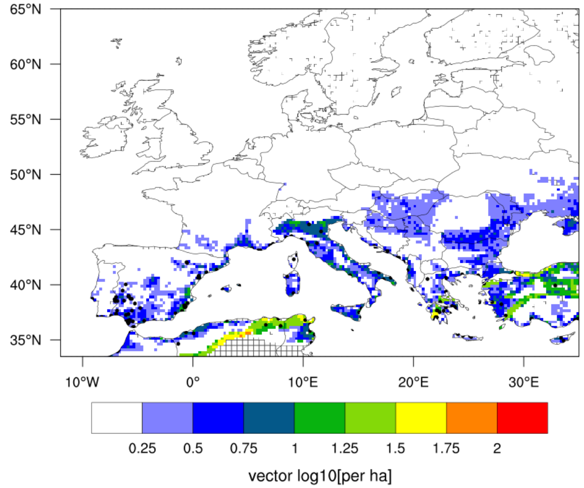
Model 1 - Kraemer et al., 2015



Model 2 – Tann > 10ºC



Model 3 - Oviposition(Tº)
Yang et al. 2009



Model 4 – *Ae. aegypti* version of VECTRI (simulated adult density)

Distribution of Ae. aegypti in Europe based on different models. Rainfall and temperature data was derived from the EOBS dataset (0.25º) for 2011-2021.

Models based on different assumptions confirm that southern Europe is still climatically suitable for the establishment of *Ae. aegypti*.

Ae. aegypti in Cyprus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The invasions of *Aedes aegypti* and *Aedes albopictus* in Cyprus: current situation, risk modelling and public health implications for the wider Eastern Mediterranean region

Cyril Caminade, Marlen I. Vasquez, Herodotos Herodotou, Gregoris Notarides, Costas Pavlou, Filip Fayad, Apostolos Papakonstantinou, Marios Violaris, Dusan Petric, Wadaka Mamai, Adrian M. Tompkins, Jeremy Bouyer

doi: https://doi.org/10.1101/2024.12.17.628941

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Posted December 20, 2024.

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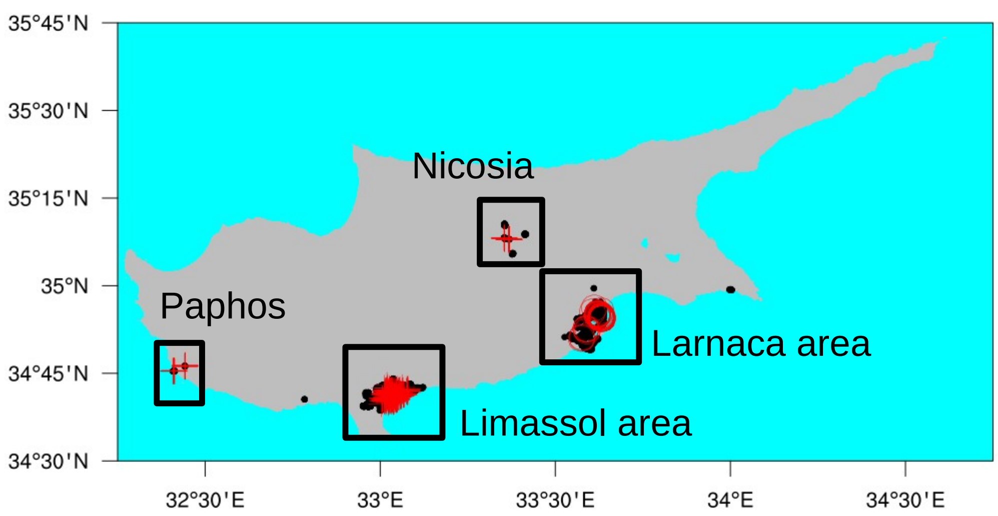


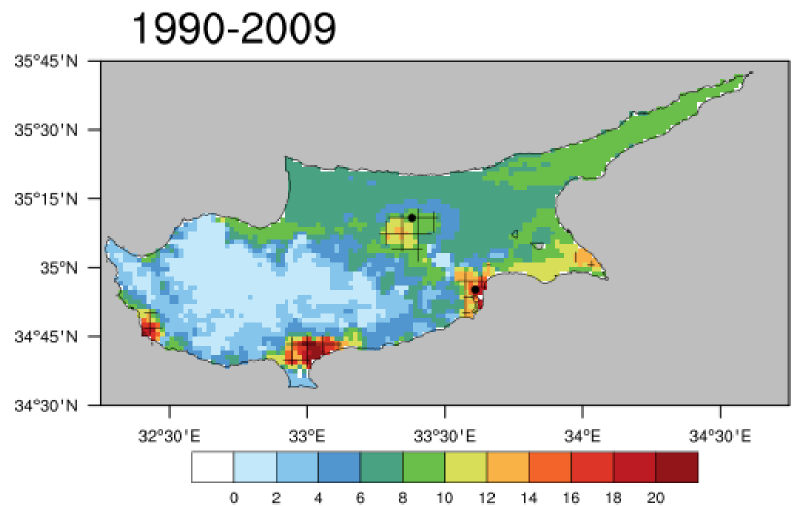
Ovitrap data collected between Oct 2022 and June 2023 over Cyprus (CIT team led by M. Vasquez & IAEA team led by J. Bouyer).

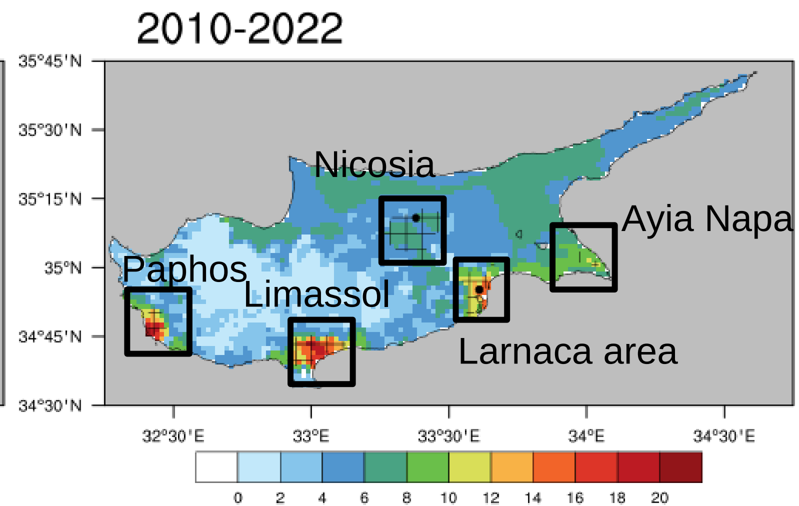
Negative traps are depicted by black dots, positive *Ae. aegypti* traps are denoted by red circles and positive *Ae. albopictus* traps are indicated by red crosses.

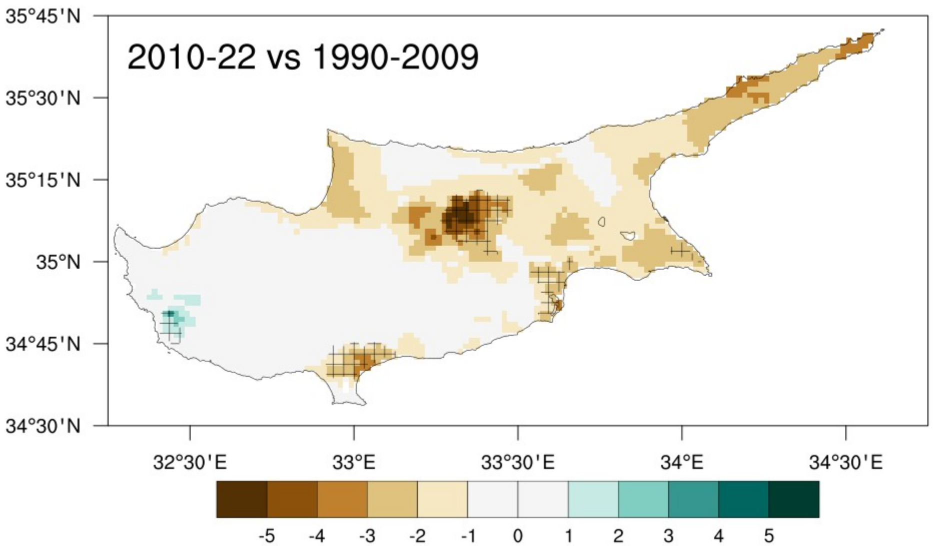
Aedes mosquitoes are present in urban areas of Cyprus.

Caminade et al. (2024)
https://doi.org/10.1101/2024.12.17.628941

Simulated distribution of *Ae. aegypti* and changes in Cyprus



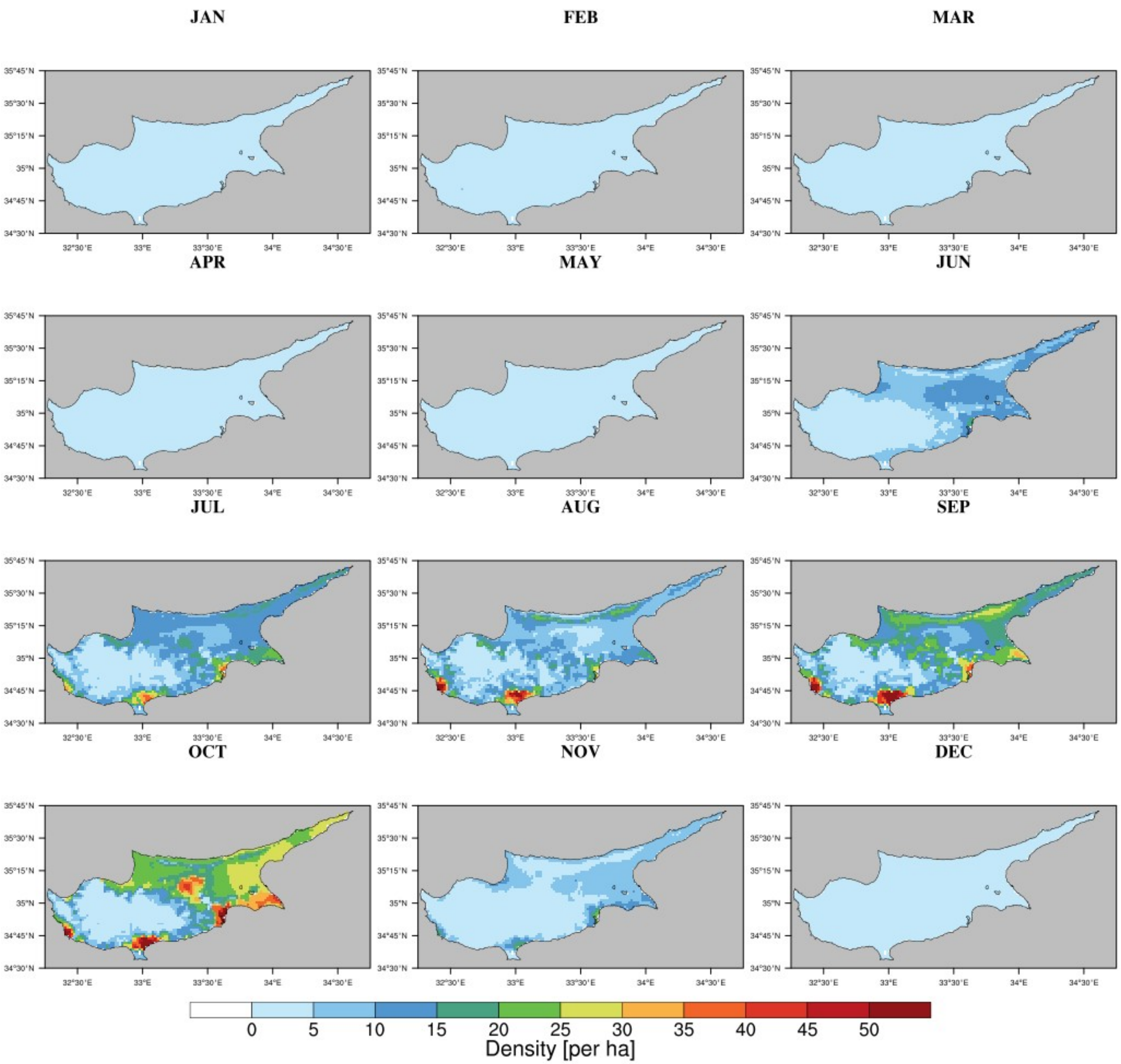




***Top**: Simulated adult Ae. aegypti density (per ha) by the VECTRI model for the periods 1990-2009 and 2010-22. Hatching denotes areas > 250 person/km2. **Bottom**: Recent changes in simulated adult density (2010-22 vs 1990-2009).*

Hotspots simulated over densely populated regions of Cyprus: Paphos, Limassol, Larnaca, Nicosia and Ayia Napa. Moderate risk simulated over the northern coasts.
Decrease in simulated abundance during the past 10-15y, excepting over Paphos.

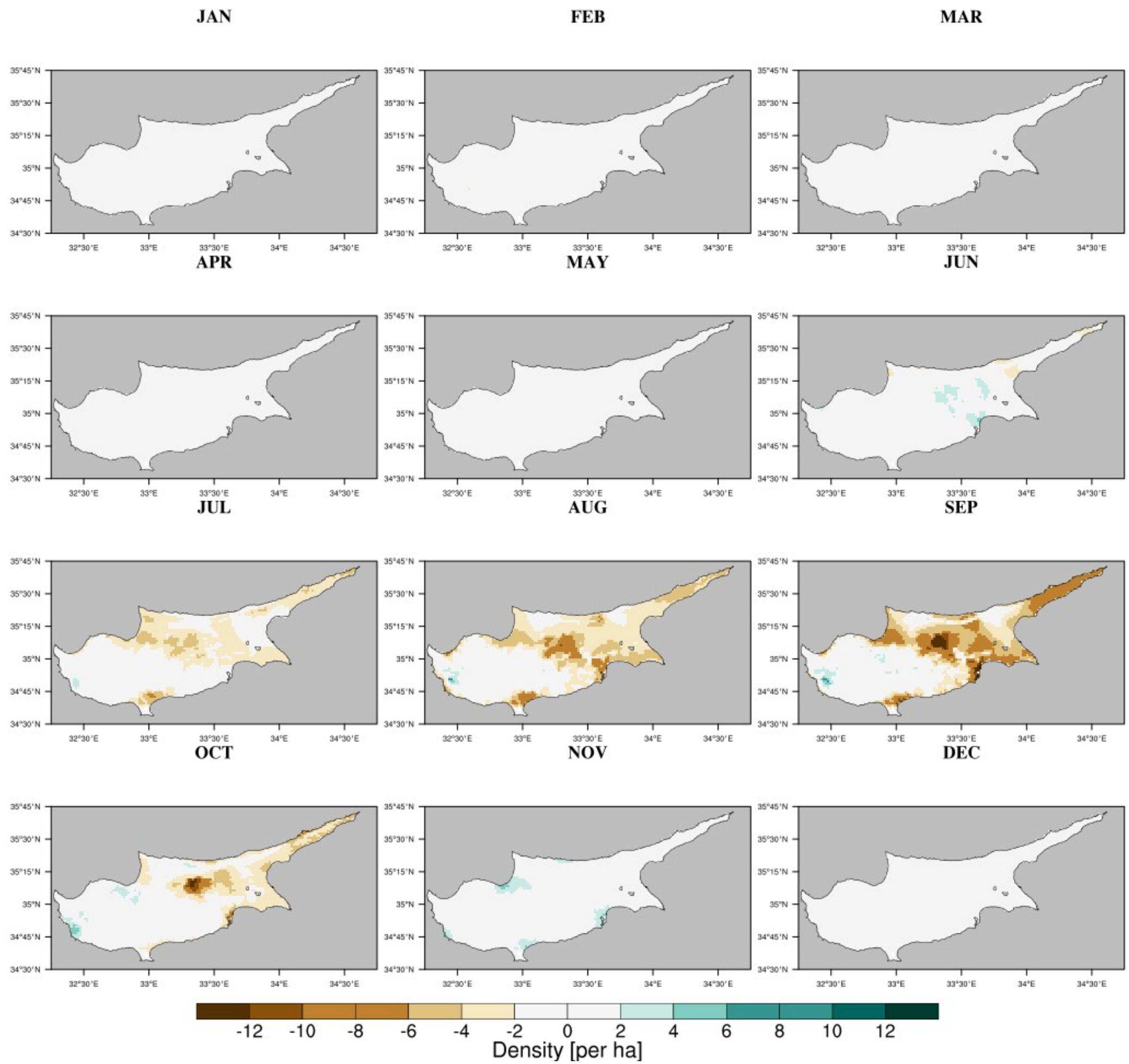
Simulated seasonal cycle in Cyprus: 1990-2022



***Left**: Mean seasonal cycle of simulated adult Ae. aegypti density (per ha) by the VECTRI model for 1990-2022.*

Simulated mosquito activity between June and November with peaks in Sep-Oct over Cyprus.

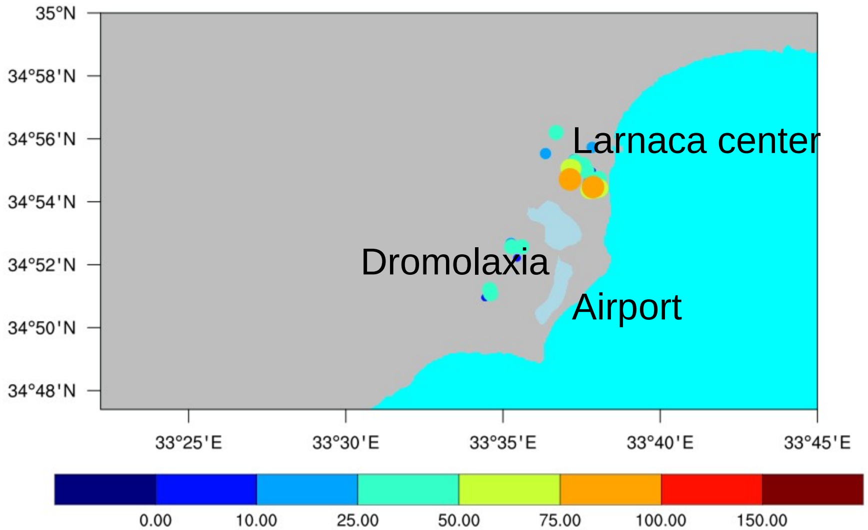
Simulated seasonal cycle in Cyprus: 2010-22 vs 1990-2009

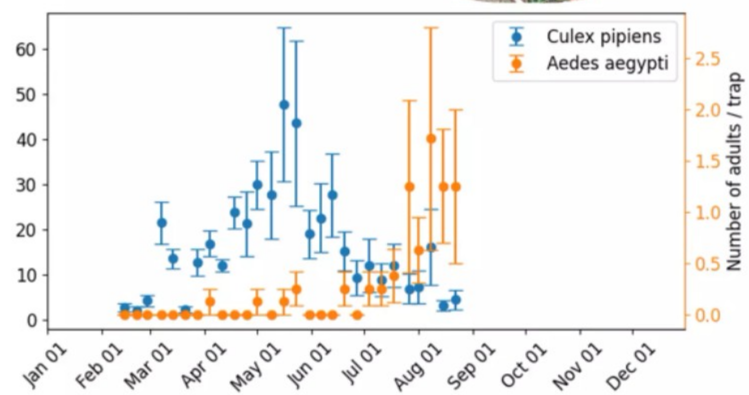


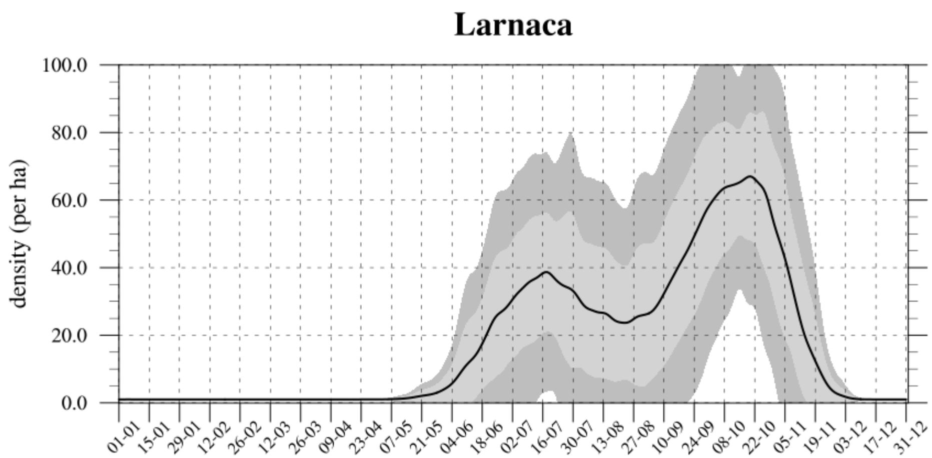
***Left**: Changes in seasonal cycle for simulated Ae. aegypti density (per ha) between 2020-22 and 1990-2009.*

Simulated decrease in abundance during the warmest months (Aug-Sep), in particular over Nicosia. Slight increase in climate suitability in June and November.

Ae. aegypti in Larnaca





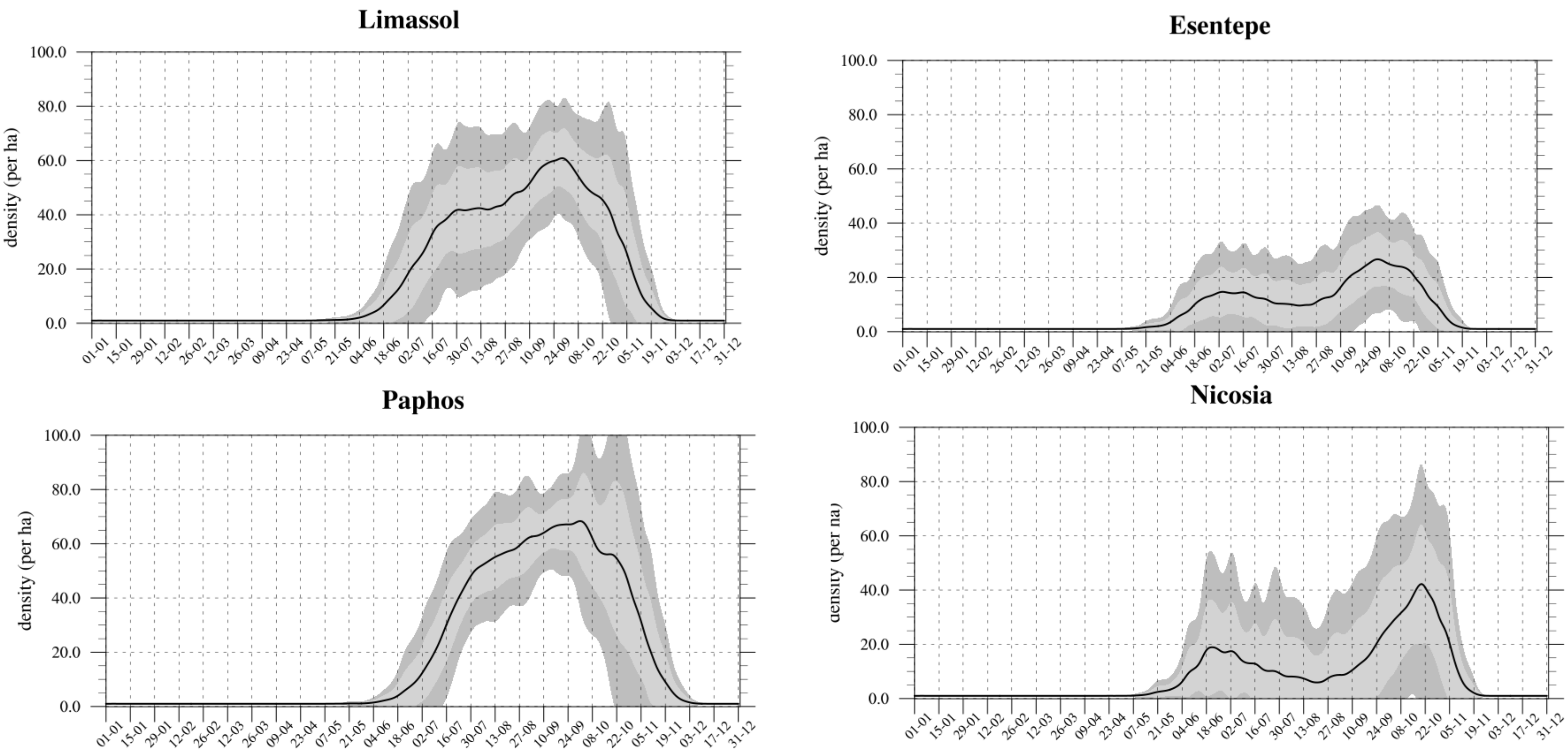


***Top**: Averaged egg abundance (per trap per week) for Ae. aegypti in Larnaca area for the period 24 Oct – 5 Dec 2022.*

***Top right**: Number of adult Ae. aegypti in Larnaca (BGS trap data).*

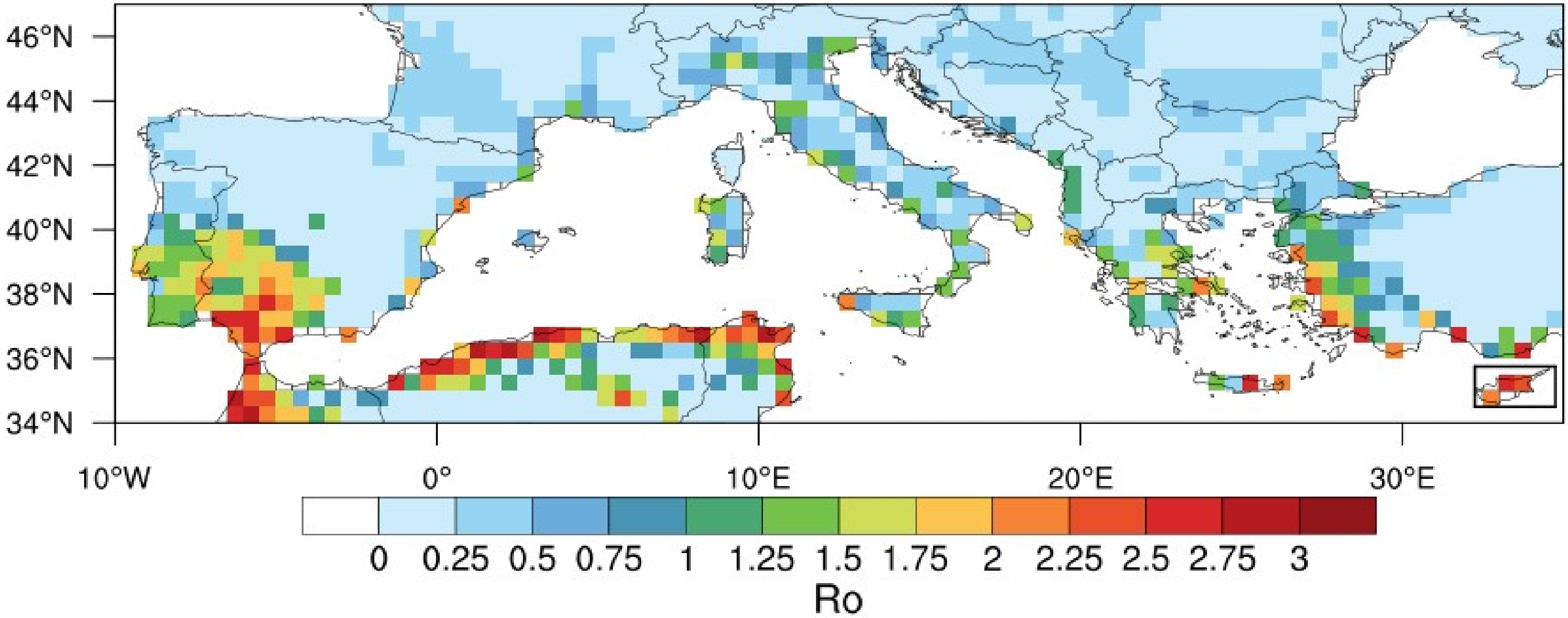
***Bottom right**: Simulated adult density (per ha) over Larnaca for the period 1990-2022.*

Simulated seasonal cycles of Ae. aegypti in cities



Mean seasonal cycle of simulated larval density for Ae. aegypti in Limassol, Esentepe, Paphos and Nicosia. The mean for 1990-2022 is depicted by the black solid line. The light-gray and gray envelope respectively show one and two standard deviations around the mean calculated for all years.

Risk of arbovirus transmission over the MED region



Annual mean basic reproduction number (R_0) for the risk of arbovirus transmission by Ae. aegypti & Ae. albopictus over the Mediterranean region (1980-2010). Cyprus is highlighted by the black box. One host – two vectors (Ae. albopictus and Ae. aegypti) R_0 model developed for ZIKV (Caminade et al., PNAS 2017).

These hotpots are consistent with **historical** dengue **epidemics** reported in **Cyprus** in 1861, 1888-89, 1913 and 1928; in **Crete** in 1881; over **western Greece** in 1889, 1895-97,1910, 1927-28, 1929-33 ; in **Napoli** in 1889-90, in **Turkey** in 1889-90, 1899, 1916, 1927-28 and 1945 and over **Andalucía** in **Spain** in 1784-88-93, 1863-67, 1887 and 1928 (Schaffner & Mathis, Lancet Inf. Dis. 2014).

Largest documented dengue epidemic in Athens in 1927-28 (Louis C., Emerg Infect Dis. 2012).

Summary

- Statistical and dynamical models reproduce the historical distribution of *Ae. aegypti* over Europe. *Ae. aegypti* was present over the southern coasts of the Mediterranean, along the coasts of the Adriatic, and northern Africa.
- The VECTRI models simulate “climatic hotspots” over the urban areas of Larnaca, Limassol, Paphos, Ayia Napa, Nicosia and to a lesser extent the northern coasts. Recent climatic conditions less suitable over urban areas due to high temperature conditions over summer.
- Bimodal activity of *Ae. aegypti* for Larnaca, Nicosia and Esentepe (with a peak in June-July then October). Recent climate change tends to increase simulated mosquito density over the season edges and decrease during the warmest months in summer.
- Historical transmission of dengue and Yellow Fever occurred in main ports of the Mediterranean before vector/malaria control measures were undertaken post WWII.
- But *Ae. aegypti* has failed to re-establish in southern Europe. Other reasons might prevent its re-establishment (competition with other mosquito species? piped water? Lower potential to overwinter? Transportation by boat and/or aircraft?). See *Wint et al. (2022)*

Recommendations

- Stringent and continuous surveillance effort (deployment of ovitrap and BGS) should be developed and intensified in Larnaca but also Limassol, Paphos, Ayia Napa, Nicosia, and over the northern coast and eastern part of Cyprus (work in progress).
- Stringent surveillance of imported dengue, Zika and chikungunya cases in spring-summer in portal cities and at Larnaca international airport should be conducted to avoid potential autochthonous disease cases occurring during early summer and fall.
- Published epidemiological study found that “the introduction of dengue cases to Madeira in 2012 occurred around a month before first autochthonous cases were reported, during the maximum influx of airline travellers, and that declining temperature conditions in autumn were critical in limiting the lifespan of the dengue outbreak in early December 2012” (Lourenco et al., 2014) -> Learn from past epidemics
- Given that Greece is the main trading partner of Food products into Cyprus (World Bank, 2021), stringent surveillance should be conducted in containers transiting between the main harbours of both countries and in large airports.
- Sterile Insect Technique and other vector control approaches to eradicate *Ae. aegypti* in Cyprus.

- Model comparison exercise required to estimate uncertainties and provide science-informed decision making.

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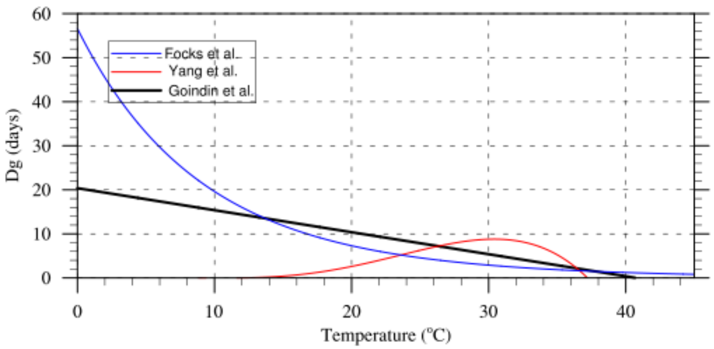
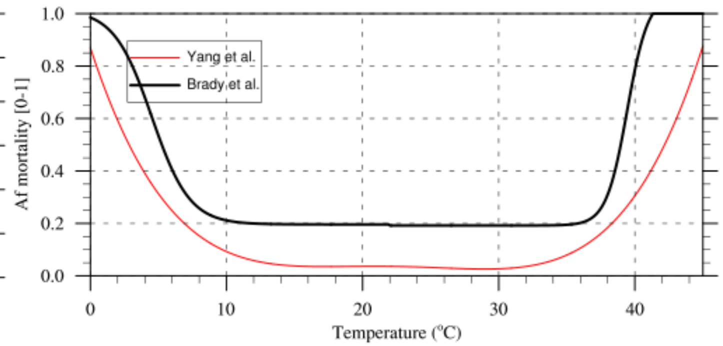
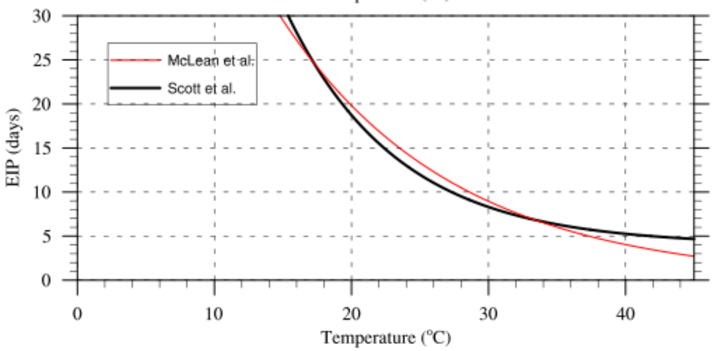
Joint FAO/IAEA Division of Nuclear Techniques in Food and Agriculture

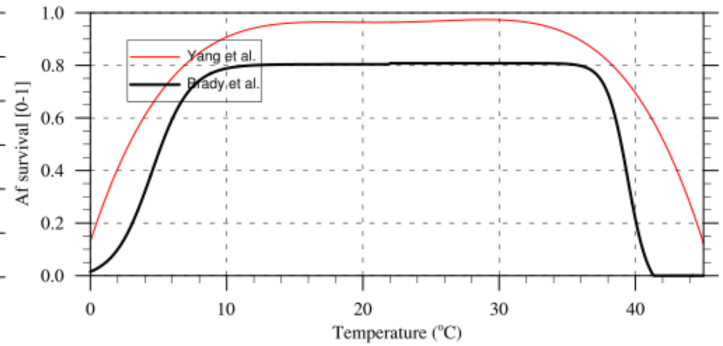
Extra slides

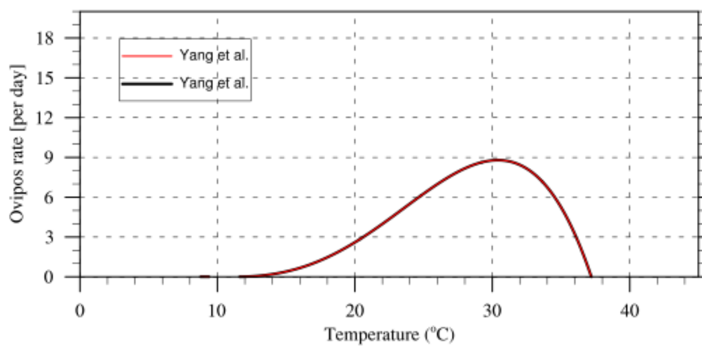
Data & Methods











- Yang et al., 2009 [https://doi.org/10.1017/S0950268809002040]
- Brady et al., 2013 [https://doi.org/10.1186/1756-3305-6-351]
- McLean et al., 1974 [https://doi.org/10.1139/m74-040]
- Focks et al., 1993 [https://doi.org/10.1093/jmedent/30.6.1003]
- Goindin et al., 2015 [https://doi.org/10.1371/journal.pone.0135489]
- Scott et al., 2000 [https://doi.org/10.1603/0022-2585-37.1.89]

- EIP(T) for dengue virus
- Af: Adult females

Ae. aegypti in Larnaca - animation